TUTORIAL XVII

Q1. We wish to fit a polynomial of the form $y = \beta_0 + \beta_1 x + \beta_2 x^2$ (which represents a parabola) to the 10 points as given in the table (Reactions to two drugs):

i	1	2	3	4	5	6	7	8	9	10
x_i	1.9	0.8	1.1	0.1	−0.1	4.4	4.6	1.6	5.5	3.4
y_i	0.7	−1.0	−0.2	−1.2	−0.1	3.4	0.0	0.8	3.7	2.0

Obtain normal equations.

SOLUTION

Tutorial XVII

Q1. The quadratic polynomial is

$$y = \beta_0 + \beta_1 x + \beta_2 x^2 \qquad (1)$$

Choose β's s.t.

$$S = \sum_{i=1}^{10} (y_i - \beta_0 - \beta_1 x - \beta_2 x^2)^2 \qquad (2)$$

is minimum. Setting $\frac{\partial S}{\partial \beta_0}$, $\frac{\partial S}{\partial \beta_1}$, $\frac{\partial S}{\partial \beta_2}$ equal to zero, we get <u>normal eqns</u>

$$10\beta_0 + \beta_1 \Sigma x_i + \beta_2 \Sigma x_i^2 = \Sigma y_i$$

$$\beta_0 \Sigma x_i + \beta_1 \Sigma x_i^2 + \beta_2 \Sigma x_i^3 = \Sigma x_i y_i \qquad (3)$$

$$\beta_0 \Sigma x_i^2 + \beta_1 \Sigma x_i^3 + \beta_2 \Sigma x_i^4 = \Sigma x_i^2 y_i$$

Substituting we have

$$10\beta_0 + 23.3\beta_1 + 90.37\beta_2 = 8.1$$

$$23.3\beta_0 + 90.37\beta_1 + 401.0\beta_2 = 43.59$$

$$90.37\beta_0 + 401.0\beta_1 + 1892.7\beta_2 = 204.55$$

Q. 2. Suppose that both the least squares line and the least squares parabola were fitted to the same set of points. Explain why the sum of the squares of deviations of the points from the parabola cannot be larger than the sum of the squares of the deviations of the points from the straight line.

Q 2. Sum of squares of deviations from the 'least squares parabola' is

$$S_1 = \sum_{i=1}^{n} \left(y_i - \beta_0 - \beta_1 x - \beta_2 x^2\right)^2 \qquad (1)$$

where β_0, β_1 and β_2 are obtained from normal eqns and β_0, β_1 and β_2 vary over all real numbers.

When $\beta_2 = 0$, then sum of squares of deviations from the least squares straight line is

$$S_2 = \sum_{i=1}^{n} \left(y_i - \beta_0 - \beta_1 x\right)^2 \qquad (2)$$

Note that S_1 cannot be smaller than S_2.

Q3. Show that the least squares line

$$y = \hat{\beta}_0 + \hat{\beta}_1 x$$

passes through $(\bar{x}, \bar{y})$.

Q3.

$$S = \sum_{i=1}^{n} (y_i - b_0 - b_1 x_i)^2$$

Determine b_0 and b_1 so that S is minimum.

Accordingly, partial derivatives of S w.r to b_0 and b_1 should vanish separately

$$\frac{\partial S}{\partial b_0} = 0 = -2 \sum (y_i - b_0 - b_1 x_i)$$

or $\quad \sum y_i = n b_0 + b_1 \sum x_i$

or $\quad \bar{y} = b_0 + b_1 \bar{x}$

Thus the least squares line passes through $(\bar{x}, \bar{y})$.

Q4. Grosch's law suggests that CPU price y is roughly proportional to x (MAPS). We use a general power function

$$y = ax^b$$

How will you obtain the least squares fit? Comment on the 'transformed data'.

Q4. <u>Solved Example Trivedi</u>

$$\text{Min } S = \sum \left(y_i - a x_i^b\right)^2$$

(1) Choose a and b so that S is minimized. This is referred to as <u>nonlinear estimation</u> of a and b.

(2) Transformation of data by using

$$\ln Y = \ln a + b \ln X$$

does not <u>preserve</u> 'least squares property'

$$\text{Min. } \sum \left[\ln y_i - (\ln a + b \ln x_i)\right]^2$$